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11.3.1 Jacobi rotation

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turns out that you can find a rotation, Sigma -- sorry -- gamma, minus sigma, sigma, gamma, transpose. And then apply the same rotation from the other side in such a way that the result of this is a diagonal matrix. Now obviously what this does is it computes the Spectral Decomposition of a two-by-two symmetric matrix. Okay? And we know that that has to be diagonal. Well, I have a little homework where you

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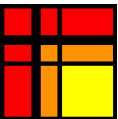
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Reading assignment

1/1 point (graded)



Read Unit 11.3.1 of the notes. [[LINK](#)]

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✔ Confusion about how to create the Jacobi Rotation Matrix	6
? Unitary Matrices In this course we do lots of multiplying with unitary matrices. Jacobi, Givens...	2

Homework 11.3.1.1

1/1 point (graded)

With Matlab, use the `eig` function to explore the eigenvalues and eigenvectors of various symmetric matrices:

```
[ Q, Lambda ] = eig( A )
```

Try, for example

```
A = [
-1  2
 2  3
]
```

```
A = [
 2 -1
-1 -2
]
```

How does the matrix Q relate to a Jacobi rotation? How would Q need to be altered for it to be a Jacobi rotation?

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